

2. Meselson and Stahl investigated the replication of DNA prior to cell division.

In their experiments, bacteria were cultured in a medium containing only the ^{15}N isotope (heavy nitrogen). The bacteria were then transferred to ^{14}N (light nitrogen) culture medium and allowed to divide. Following each generation, samples of DNA were taken, separated using an ultracentrifuge and analysed.

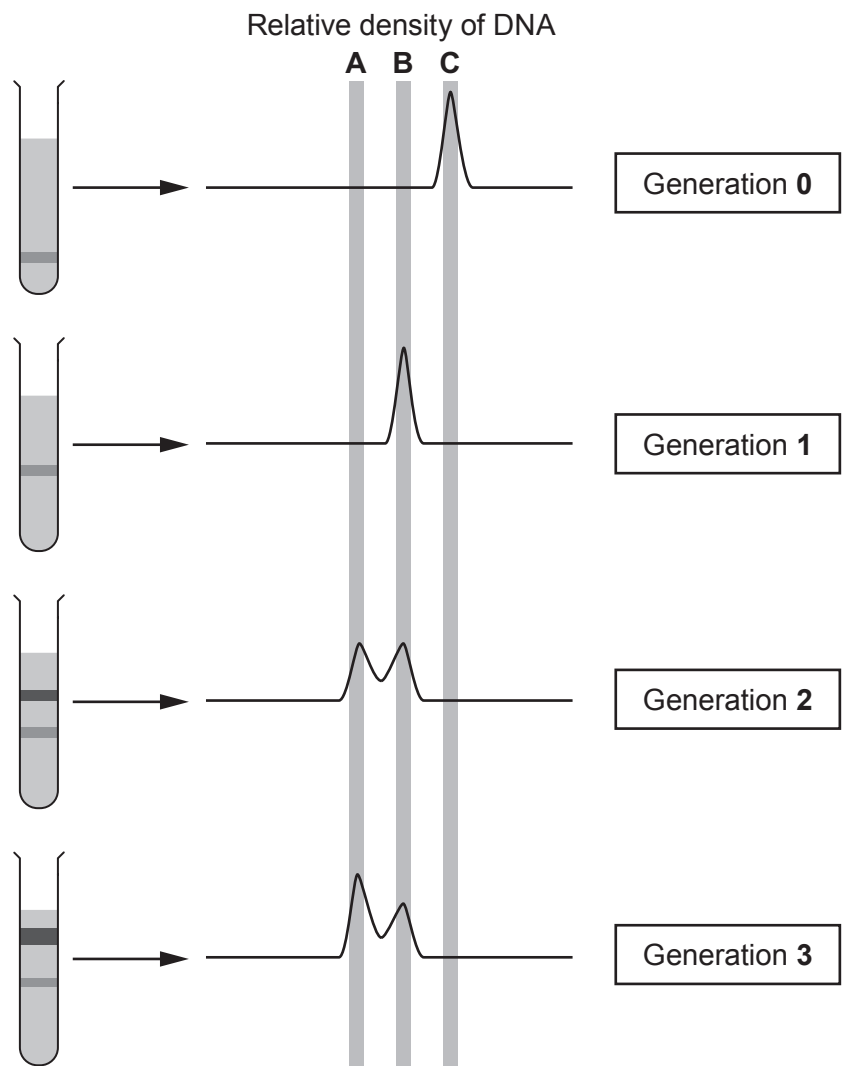
- (a) Identify **three** variables which would need to be controlled during centrifugation of the samples. [2]

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Photographs were taken of the extracted bands of DNA. The relative density of each band was scanned and measured; the more dense the DNA, the higher the reading. Shown below are some of their experimental results.



(b) (i) State which isotopes of nitrogen would be found in the DNA at **A**, **B** and **C** in the scans. [1]

A

B

C

(ii) Explain how the results in generations **0** and **1** support the theory of semi-conservative replication. [3]

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(iii) Explain the height of the peaks in the scans for generations **2** and **3**. [4]

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(iv) Name **two** enzymes now known to be involved in DNA replication and explain their functions. [3]

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